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THE DEVELOPMENT OF AFFECTIONAL RESPONSES IN INFANT MONKEYS[1]

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(*Read April 24, 1958*)

HUMAN beings have long observed the development of affection or love in the neonate and infant. It is an obvious fact that this early affection of the infant is first intimately bound to the mother and then gradually expands, through mechanisms by no means well known, to other human beings.

Observational evidence, often controlled, orderly, and systematic, has led scientists to speculate both on the nature of the early affectional responses and the development and generalization of these responses. Although experimental psychologists have given little attention to the fundamental nature or development of affection, their common theoretical position is that affection is a derived drive developed through the repeated association of the mother with reduction of the primary biological drives, particularly hunger and thirst.

A wide diversity of theoretical positions has been held by various psychoanalysts. Some, such as Klein (3), have emphasized the breast as an object for which the infant innately strives and connects, whereas others, including Winnecoot (7) and Ribble (5), have described broader innate needs relating to contact, movement, and temperature. Bowlby [2] has given approximately equal emphasis to primary clinging (contact) and sucking as innate affectional components, and at a later maturational level, visual and auditory following.

The role of visual and auditory following (imprinting) has been stressed by Lorenz (4) and other ethologists as primary innate mechanisms binding the infant bird and fish to the mother. Even if similar mechanisms exist in the primate, they appear at a later developmental stage and play less important roles than they do in lower animals.

There are many reasons for the absence of experimental analysis of the nature and development of affectional responses in the human infant. His motor capabilities are so limited at birth that there are few, if any, precise responses that can be measured and recorded, and by the time his precise responses can be measured, the antecedent determining variables have been obscured in a complex maze of confounded variables previously unmeasurable.

Fortunately, these difficulties are resolved in large part if we use the neonatal and infant macaque monkey as the subject for the experimental analysis of basic affectional variables. It is possible to make precise measurements in this primate beginning at two to ten days of age, depending upon the maturational status of the individual subject at birth. The macaque infant differs from the human infant in that the monkey is more mature at birth and grows more rapidly, but the basic responses relating to affection, including nursing, contact, clinging, and even visual and auditory exploration, exhibit no fundamental differences in the two species. Even the development of perception, emotion, and learning capability follow very similar sequences.

In an effort to gain complete control over the lives of the monkeys used as subjects for affectional analysis, we separated these animals from their mothers six to twelve hours after birth and suckled them on tiny nursing bottles. Three years' previous experience with some sixty neonates, also separated from their mothers at birth, had demonstrated the complete feasibility of such procedures, for these earlier laboratory-raised monkeys were larger and healthier and had a higher incidence of survival than infants left to the care of their monkey mothers.

We had noted that the previous laboratory-raised babies showed strong attachment to the cloth pads (folded gauze diapers) which were used to cover the hardware-cloth floors of their cages. The infants clung to these pads and engaged in violent temper tantrums when the pads

[1] Support for the research presented in this paper was provided through funds received from the Graduate School of the University of Wisconsin, and from Grant M-772, National Institutes of Health, and from a Ford Foundation Grant.

[2] Personal communication from Dr. John Bowlby, Tavistock Clinic, London, England.

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FIG. 1. Response to cloth pad by one-day-old monkey.

were removed and replaced for sanitary reasons. Such contact-need or responsiveness had been reported previously by Van Wagenen (6) and is reminiscent of the devotion often exhibited by human infants to their pillows, blankets, and soft, cuddly stuffed toys. Responsiveness by the one-day-old infant monkey to the cloth pad is shown in figure 1, and an unusual and strong attachment of a six-month-old infant to the cloth pad is illustrated in figure 2.

We took these observational data as the cue for the construction of two mother surrogates, one designed to be soft, the other to be rigid and unyielding. Figure 3 shows the two mothers. On the right is a cloth mother made from a block of wood covered by sponge rubber and sheathed with terry cloth. The wire mother on the left is made of hardware cloth, a substance entirely adequate to provide support and nursing capability. We presumed that the cloth and wire mothers would differ to the neonatal monkeys in no way other than the difference in contact comfort provided by the mother's body. Although we deliberately designed different faces for the two classes of mothers, responsiveness to the face matures at a later age than affectional responding.

FIG. 2. Response to gauze pad by six-month-old monkey used in earlier study.

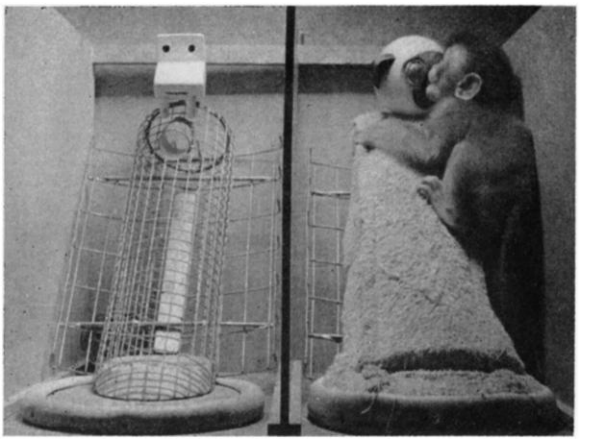

FIG. 3. Wire and cloth mother surrogates.

In our initial experiment, the dual mother-surrogate condition, a cloth mother and a wire mother were placed in different cubicles attached to the infant's living cage (see fig. 3). For four infant monkeys the cloth mother lactated and the wire mother did not, and for the other four this condition was reversed. In either condition the infant received all its milk through the mother surrogate as soon as it was able to maintain itself in this way. In the first few days, supplementary feedings were needed, but these could be eliminated after two or three days except in the case of very immature animals. Thus, the experiment was designed as a test of the relative importance of the variables of contact comfort and nursing comfort. During the first fourteen days of life the monkey's cage floor was covered with a heating pad wrapped in a folded gauze diaper, and thereafter the cage floor was bare. The infant subjects were always free to leave the heating pad or cage floor to contact either mother, and the time spent on the surrogate mothers was automatically recorded. Figure 4 shows the total time spent on the cloth and wire mothers under the two conditions of

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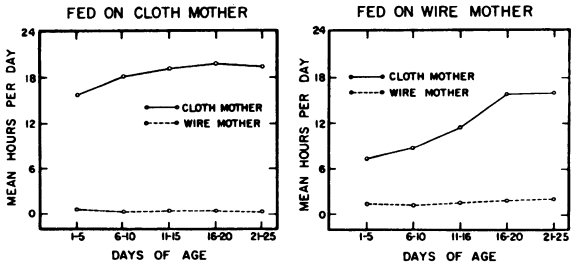

FIG. 4. Time spent on cloth and wire mother surrogates.

feeding. These data make it obvious that contact comfort is a variable of overwhelming importance in the development of affectional responses, whereas lactation is a variable of negligible importance. With age and opportunity to learn, responsiveness to the lactating wire mother decreases and responsiveness to the nonlactating cloth mother increases, a finding completely contrary to any interpretation of derived drive in which the mother-form becomes conditioned to hunger-thirst reduction. The persistence of these differential responses throughout 165 consecutive days of testing is evident in figure 5.

We were not surprised to discover that contact comfort was an important basic affectional or love variable, but we did not expect it to overshadow so completely the variable of nursing; indeed, the disparity is so great as to suggest that the primary function of nursing as an affectional variable is that of insuring frequent and intimate body contact of the infant with the mother.

One function of the real mother, human or subhuman, and presumably of a mother surrogate, is to provide a haven of safety for the infant in times of fear and danger. The frightened or ailing child clings to its mother, not its father, and this selective responsiveness in times of distress, disturbance, or danger may be used as a measure of the strength of affectional bonds. We have tested this kind of differential responsiveness by presenting to the infants in their cages, in the presence of the two mothers, various fear-producing stimuli such as the moving toy bear illustrated in figure 6. The data on differential responsiveness are presented in figure 7, and it is apparent that the cloth mother is highly preferred over the wire one, and this differential selectivity is enhanced by age and experience. In this situation, the variable of nursing appears to be of absolutely no importance; the infant consistently seeks the soft mother surrogate regardless of nursing condition.

Similarly, the mother or mother surrogate provides its young with a source of security, and this role or function is seen with special clarity when mother and child are in a strange situation.

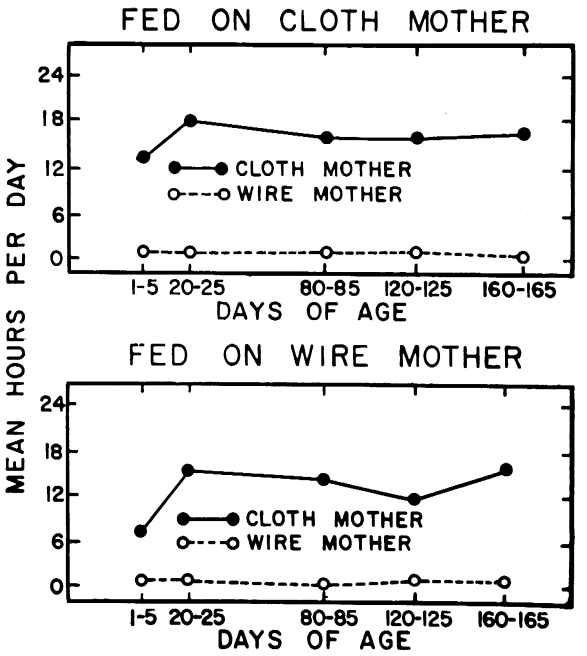

FIG. 5. Long-term contact time on cloth and wire mother surrogates.

FIG. 6. Typical fear stimulus.

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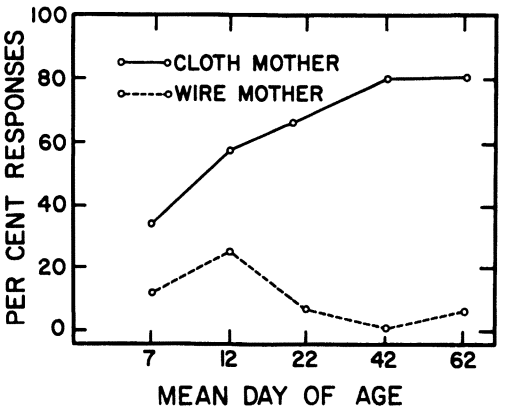

FIG. 7. Differential responses to the two mothers in fear tests.

At the present time we have completed tests for this relationship on four of our eight baby monkeys assigned to the dual mother-surrogate condition by introducing them for three minutes into the strange environment of a room measuring six feet by six feet by six feet (also called the "open-field test") and containing multiple stimuli known to elicit curiosity-manipulatory responses in baby monkeys. The subjects were placed in this situation twice a week for eight weeks with no mother surrogate present during alternate sessions and the cloth mother present during the others. A cloth diaper was always available as one of the stimuli throughout all sessions. After one or two adaptation sessions, the infants always rushed to the mother surrogate when she was present and clutched her (see fig. 8), a response so strong that it can be adequately depicted only by motion pictures. After a few additional sessions, the infants began to use the mother surrogate as a source of

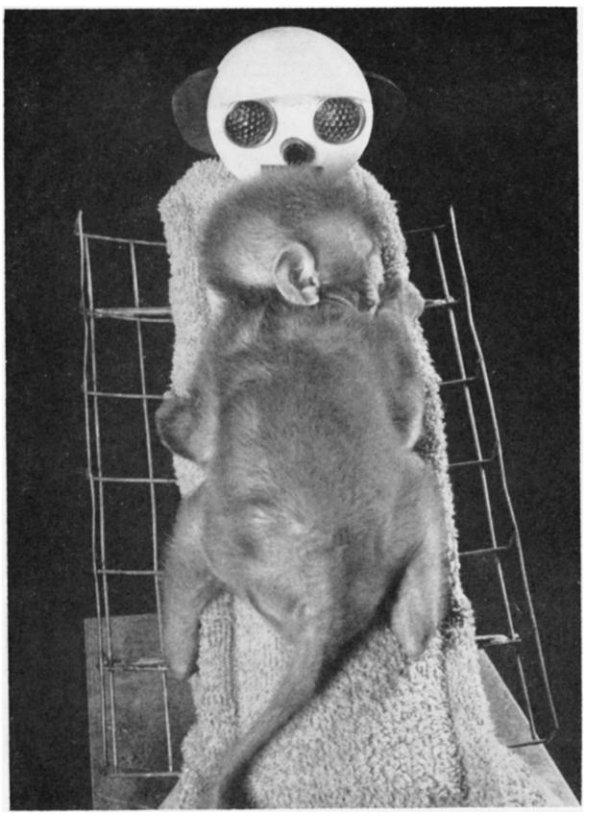

FIG. 8. Initial response to cloth mother in the open-field tests.

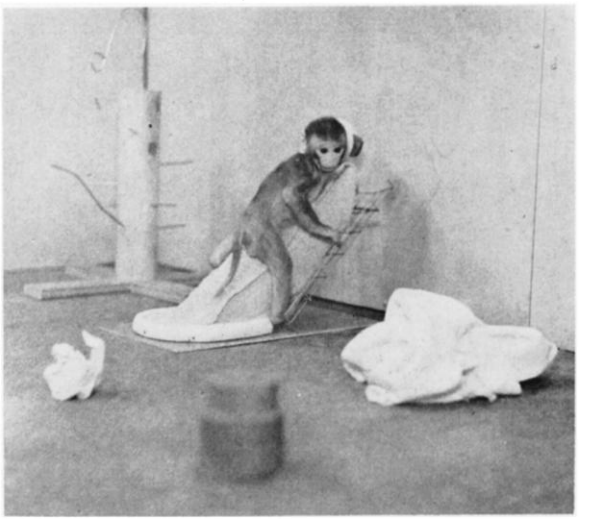

FIG. 9. Subsequent response to cloth mother in the open-field tests.

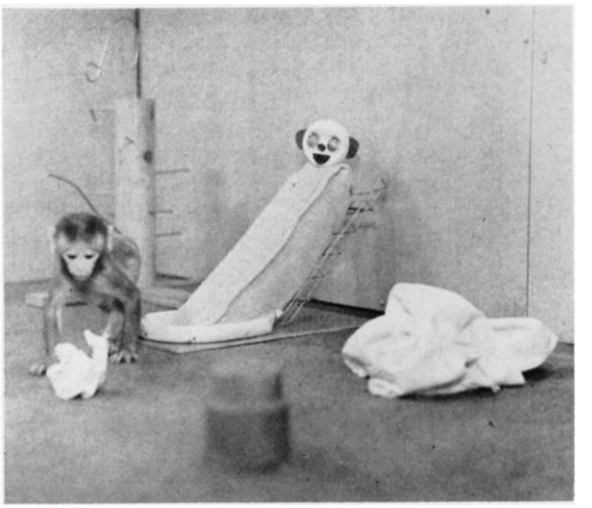

FIG. 10. Object exploration in presence of cloth mother.

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security, a base of operations. As is shown in figures 9 and 10, they would explore and manipulate a stimulus and then return to the mother before adventuring again into the strange new world. The behavior of these infants was quite different when the mother was absent from the room. Frequently they would freeze in a crouched position, as is illustrated in figures 11*a* and 11*b*. Emotionality indices such as vocalization, crouching, rocking, and sucking increased sharply, as shown in figure 12. Total emotionality score was cut in half when the mother was present. In the absence of the mother some of the experimental monkeys would rush to the center of the room where the mother was customarily placed, and then run rapidly from object to object screaming and crying all the while. Continuous, frantic clutching of

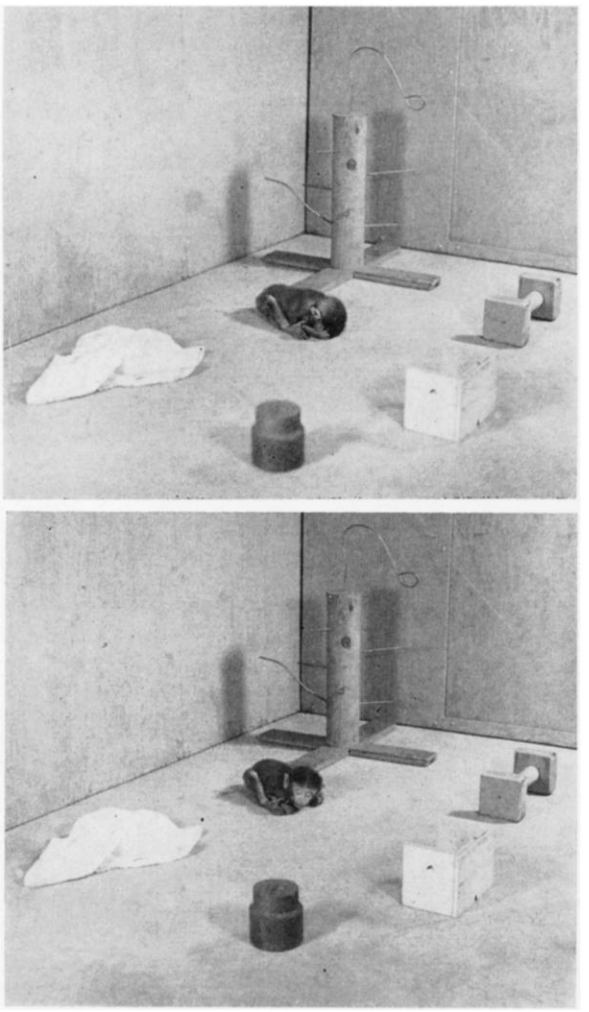

FIG. 11. *a* & *b*. Responses in the open-field test in the absence of the mother surrogate.

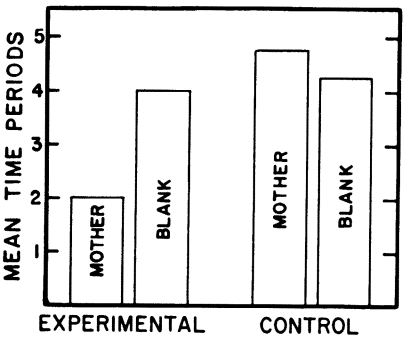

FIG. 12. Emotionality index with and without the presence of the cloth mother.

their bodies was very common, even when not in the crouching position. These monkeys frequently contacted and clutched the cloth diaper, but this action never pacified them. No difference between the cloth-mother-fed and wire-mother-fed infants was demonstrated under either condition. Four control infants never raised with a mother surrogate showed the same emotionality scores when the mother was absent as the experimental infants showed in the absence of the mother, but the controls' scores were slightly larger in the presence of the mother surrogate than in her absence.

Some years ago Butler (1) showed that mature monkeys enclosed in a dimly lighted box would open and reopen a door for hours on end for no other motivation than that of looking outside the box. We now have data demonstrating that neonatal monkeys show this same compulsive visual curiosity on their first test day in an adaptation of the Butler apparatus. Usually these tests are begun at ten days of age, but this same persistent visual exploration has been obtained in a three-day-old monkey during the first half-hour of testing. Butler (2) also demonstrated that rhesus monkeys show selectivity in rate and frequency of door-opening to stimuli of differential attractiveness in the visual field outside the box. We have utilized this principle of response selectivity by the monkey to measure strength of affectional responsiveness in our infants in the baby version of the Butler box. The test sequence involves four repetitions of a test battery in which the four stimuli of cloth mother, wire mother, infant monkey, and empty box are presented for a thirty-minute period on successive days. The first four subjects in the dual mother-surrogate group were given a single test sequence at forty to fifty days of age, depending upon the availability of the apparatus, and only their data are presented.

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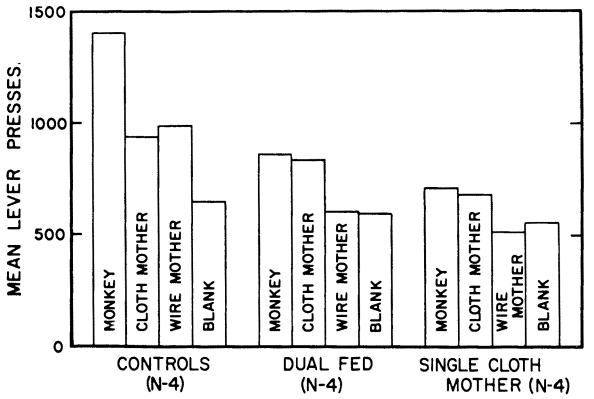

FIG. 13. Differential responses to visual exploration.

The second group of four subjects is being given repetitive tests to obtain information relating to the development of visual exploration. The data obtained from the first four infants raised with the two mother surrogates are presented in the middle graph of figure 13 and show approximately equal responding to the cloth mother and another infant monkey, and no greater responsiveness to the wire mother than to an empty box. Again, the results are independent of the kind of mother that lactated, cloth or wire. The same results are found for a control group raised, but not fed, on a single cloth mother, and these data appear in the graph on the right. Contrariwise, the graph on the left shows no differential responsiveness to cloth and wire mothers by a second control group, which was not raised on any mother surrogate.

The first four infant monkeys in the dual mother-surrogate group were separated from their mothers between 165 and 170 days of age and tested for retention during the following nine days and then at thirty-day intervals for five successive months. Affectional retention as measured by the modified Butler box is given in figure 14. In keeping with the data obtained on adult monkeys by Butler, we find a high rate of responding to any stimulus, even the empty box. But throughout the entire 160-day retention period there is a consistent and significant difference in response frequency to the cloth mother contrasted with either the wire mother or the empty box, and no consistent difference between wire mother and empty box.

Affectional retention was also tested in the open field during the first nine days after separation and then at thirty-day intervals, and each test condition was run twice at each retention interval. The infant's behavior differed from that observed during the period preceding separation. When the cloth mother was present in the post-separation period, the babies rushed rapidly to her, climbed up her body, clung tightly to her, and rubbed their head and face about her body. After this initial embrace and reunion, they played on the mother, including biting and tearing at her cloth cover; they rarely made any attempt to leave her during the test period, nor did they manipulate or play with the objects in the room, in contrast with their behavior before maternal separation. The only exception was the occasional monkey that left the mother surrogate momentarily, grasped the folded piece of paper (one of the standard stimuli in the field), and brought it quickly back to the mother. It appears that deprivation had enhanced the tie to the mother, and the contact-comfort need had become so prepotent that it overwhelmed the exploratory motives during the brief, three-minute test sessions. No change in these behaviors was observed throughout the 160-day period. When the mother was absent from the open field, the behavior of the infants was similar in the initial retention test to that during the pre-separation tests, but they tended to show gradual adaptation to the open-field situation with repeated testing and, consequently, a reduction in their emotionality scores.

In the last five retention test periods an additional test was introduced in which the surrogate mother was placed in the center of the room and covered with a clear Plexiglas box. The monkeys were initially disturbed and frustrated when their explorations and manipulations of the box failed to provide contact with the mother. However, all animals adapted to the situation rather rapidly, and soon they used the box as a place of orienta-

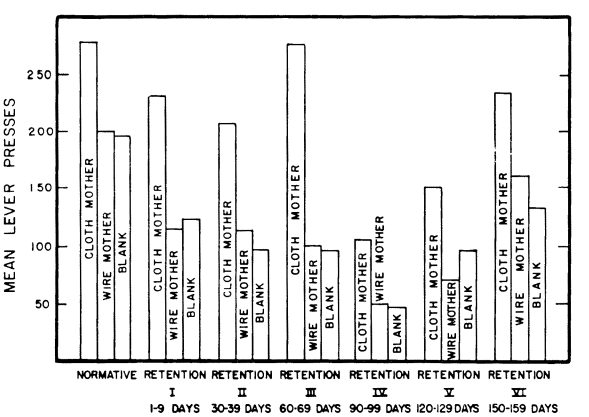

FIG. 14. Retention of differential visual-exploration responses.

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tion for exploratory and play behavior, made frequent contacts with the objects in the field, and very often brought these objects to the Plexiglas box. The emotional index was slightly lower than in the condition of the available cloth mothers, but it in no way approached the emotionality level displayed when the cloth mother was absent. Obviously, the infant monkeys gained emotional security by the presence of the mother even though contact was denied.

Affectional retention has also been measured by tests in which the monkey must unfasten a three-device mechanical puzzle to obtain entrance into a compartment containing the mother surrogate. All the trials are initiated by allowing the infant to go through a locked door, and in half the trials it finds the mother present and in half, an empty compartment. The door is then locked and a ten-minute test conducted. In tests given prior to separation from the surrogate mothers some of the infants had solved this puzzle and others had failed. The data of figure 15 show that on the last test before separation there were no differences in total manipulation under mother-present and mother-absent conditions, but striking differences exist between the two conditions throughout the post-separation test periods. Even though there is an apparent trend for total amount of manipulation to decrease, the ratio of responsiveness to the mother-present and mother-absent conditions shows no sign of decreasing. Again, there is no interaction with conditions of feeding.

The over-all picture obtained from surveying the retention data is unequivocal. There is little, if any, waning of responsiveness to the mother throughout this five-month period as indicated by any measure. It becomes perfectly obvious that this affectional bond is highly resistant to forgetting and that it can be retained for very long periods of time by relatively little reinforcement of the contact variable. During the next year retention tests will be conducted at ninety-day intervals, and further plans are dependent upon the results obtained. It would appear that affectional responses may show as much resistance to extinction as has been previously demonstrated for learned fears and learned pain, and such data would be in keeping with those of common human observation.

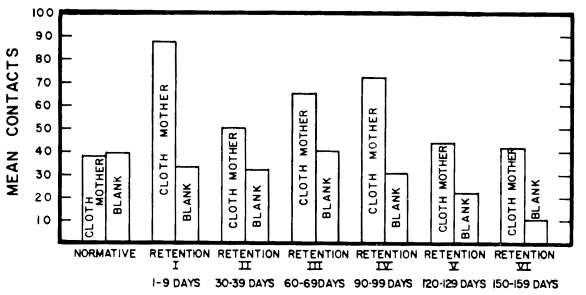

FIG. 15. Retention of puzzle manipulation responsiveness.

We have already described the group of four control infants that had never lived in the presence of any mother surrogate and had demonstrated no sign of affection or security in the presence of the cloth mothers introduced in test sessions. When these infants reached the age of 250 days, cubicles containing both a cloth mother and a wire mother were attached to their cages. There was no lactation in these mothers, for the monkeys were on a solid-food diet. The initial reaction of the monkeys to the alterations was one of extreme disturbance. All the infants screamed violently and made repeated attempts to escape from the cage whenever the door was opened. They kept a maximum distance from the mother surrogates and exhibited a considerable amount of rocking and crouching behavior, indicative of emotionality. Our first thought was that the critical period for the development of maternally directed affection had passed and that these macaque children were doomed to live as affectional orphans. Fortunately, these behaviors continued for only twelve to forty-eight hours and then gradually ebbed, changing from indifference to active contact on, and exploration of, the surrogates. The home-cage behavior of these control monkeys slowly became similar to that of the animals raised with the mother surrogates from birth. Their manipulation and play on the cloth mother became progressively more vigorous to the point of actual mutilation, particularly during the morning, after the cloth mother had been given her daily change of terry covering. The control subjects were now actively running to the cloth mother when frightened and had to be coaxed from her to be taken from the cage for formal testing.

Objective evidence of these changing behaviors is given in figure 16, which plots the amount of time these infants spent on the mother surrogates. Within ten days mean contact time is approximately nine hours, and this measure remains relatively constant throughout the next thirty days. Consistent with the results on the subjects reared from birth with dual mothers, these late-adopted infants spent less than one and one-half hours per

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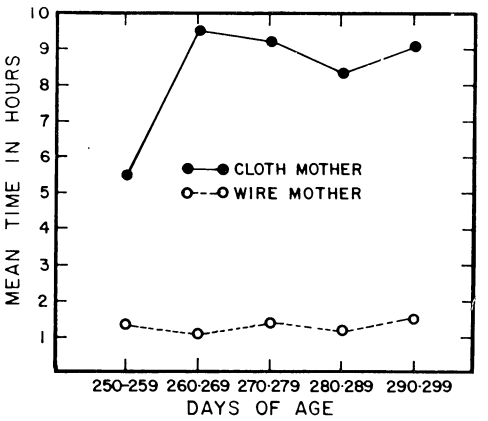

FIG. 16. Differential time spent on cloth and wire mother surrogates by monkeys started at 250 days of age.

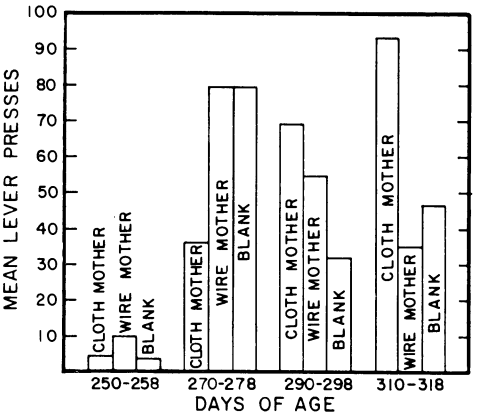

FIG. 17. Differential visual exploration of monkeys started at 250 days of age.

day in contact with the wire mothers, and this activity level was relatively constant throughout the test sessions. Although the maximum time that the control monkeys spent on the cloth mother was only about half that spent by the original dual mother-surrogate group, we cannot be sure that this discrepancy is a function of differential early experience. The control monkeys were about three months older when the mothers were attached to their cages than the experimental animals were when their mothers were removed and the retention tests begun. Thus, we do not know what the amount of contact would be for a 250-day-old animal raised from the start with surrogate mothers. Nevertheless, the magnitude of the differences and the fact that the contact-time curves for the mothered-from-birth infants had remained constant for almost 150 days suggest that early experience with the mother is a variable of measurable importance.

The control group has also been tested for differential visual exploration after the introduction of the cloth and wire mothers, and these behaviors are plotted in figure 17. By the second test session a high level of exploratory behavior has developed, and the responsiveness to the wire mother and the empty box is significantly greater than that to the cloth mother. This is probably not an artifact since there is every reason to believe that the face of the cloth mother is a fear stimulus to most monkeys that have not had extensive experience with this object during the first forty to sixty days of life. Within the third test session a sharp change in trend occurs, and the cloth mother is then more frequently viewed than the wire mother or the blank box, and this trend continues during the fourth session, producing a significant preference for the cloth mother.

Before the introduction of the mother surrogate into the home-cage situation only one of the four control monkeys had ever contacted the cloth mother in the open-field tests, and, in general, the surrogate mother not only gave the infants no security, but instead appeared to serve as a fear stimulus. The emotionality scores of these control subjects were slightly higher during the mother-present test sessions than during the mother-absent test sessions. In figure 18 the mean emotionality score for all the open-field tests given prior to the introduction of the surrogates to the home cage is compared with the mean of the fourth post-introduction test. In the absence of the cloth mothers the emotionality index in this fourth test remains near the earlier level, but the score is reduced by half when the mother is present, a result strikingly similar to that found

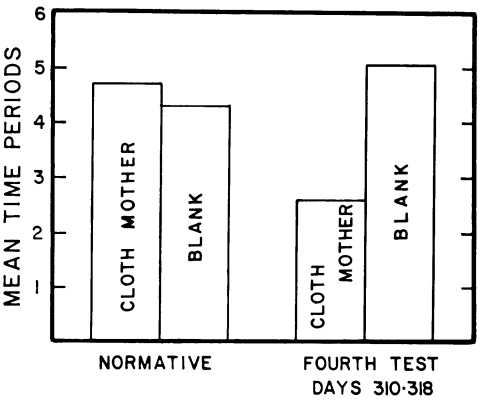

FIG. 18. Differential emotionality scores in open-field test for monkeys started at 250 days of age.

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for infants raised with the dual mother surrogates from birth. The control infants now show increasing object exploration and play behavior, and they begin to use the mother as a base of operations as did the infants raised from birth with the mother surrogates. However, there are still definite differences in the behavior of the two groups. The control infants do not rush directly to the mother and clutch her violently, but instead they go toward, and orient around, her, usually after an initial period during which they frequently show disturbed behavior, exploratory behavior, or both.

That the control monkeys develop affection or love for the cloth mother when she is introduced into the cage at 250 days of age cannot be questioned. There is every reason to believe, however, that this interval of delay depresses the intensity of the affectional response level below that of the infant monkeys that were surrogate-mothered from birth onward. In interpreting these data it is well to remember that the control monkeys had continuous opportunity to observe and hear other monkeys housed in adjacent cages and that they had had limited opportunity to view and contact surrogate mothers in the test situations, even though they did not exploit the opportunities. Finally, it should be remembered that neither surrogate was a nursing mother.

The researches which we have presented here on the analysis of affectional variables and the nature of their development in monkeys are of necessity incomplete inasmuch as the program was initiated only a year ago and the investigations cited were planned as pilot studies outlining a broad research field. In spite of the limitations certain facts stand out in sharp relief: Multiple objective techniques, reliable and with apparent face validity, can be devised to measure the infant's affectional ties. Even an appropriate inanimate mother surrogate can readily become an object loved with intensity and for a long period of time. Satisfactory body contact is an extremely strong affectional variable and may well prove to be the most important affectional variable for primates.

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